Problem 1/20 DATE_FILTER

AMAZON

Friday Filter

Identify all order records that occurred specifically on a Friday.

SAMPLE INPUT

order_id	order_date
101	2026-02-13 (Fri)
102	2026-02-14 (Sat)
103	2026-02-20 (Fri)

EXPECTED OUTPUT

order_id	day
101	Friday
103	Friday

SOLUTION

MySQL

```sql
-- Extract day name string from date
SELECT
    order_id,
    DAYNAME(order_date) as day
FROM orders
WHERE DAYNAME(order_date) = 'Friday';
```

Next: SLA Breach →

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SAVE REPOST

Problem 2/20 DATEDIFF

FEDEX

SLA Breach

Find orders where delivery took more than 7 days from the order date.

SAMPLE INPUT

order_id	ordered	delivered
205	2026-01-01	2026-01-10
206	2026-01-05	2026-01-06

EXPECTED OUTPUT

order_id	days_taken
205	9

SOLUTION

SQL Server

```sql
-- Calculate difference in days
SELECT
    order_id,
    DATEDIFF(day, ordered, delivered) as days_taken
FROM orders
WHERE DATEDIFF(day, ordered, delivered) > 7;
```

Next: Instant Conversion →

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SAVE REPOST

Problem 3/20 | INNER JOIN | META

Instant Conversion

Find users who placed an order on the exact same day they signed up.

SAMPLE INPUT

u_id	signup	order_dt
50	2024-01-01	2024-01-01
51	2024-01-02	2024-01-05

EXPECTED OUTPUT

u_id	signup	order_dt
50	2024-01-01	2024-01-01

SOLUTION

MySQL / SQL Server

```sql
-- Join on matching user AND date
SELECT
    u.user_id, u.signup, o.order_dt
FROM users u
JOIN orders o ON u.user_id = o.user_id
WHERE u.signup = o.order_dt;
```

Next: Monthly Growth →

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SAVE REPOST

Problem 4/20 | DATE_FORMAT | GOOGLE

Monthly Growth

Group and count total orders by Year-Month format.

SAMPLE INPUT

order_id	date
101	2024-01-15
102	2024-01-20
103	2024-02-05

EXPECTED OUTPUT

mo	total_orders
2024-01	2
2024-02	1

SOLUTION

MySQL

```sql
-- Convert date to 'YYYY-MM' string
SELECT
    DATE_FORMAT(date, '%Y-%m') as mo,
    COUNT(*) as total_orders
FROM orders
GROUP BY 1;
```

Next: Artist Retention →

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The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 5/20 EXISTS

SPOTIFY

Artist Retention

Find artists who released albums in both 2023 AND 2024.

SAMPLE INPUT

artist	album_yr
Drake	2023
Drake	2024
Adele	2023

EXPECTED OUTPUT

name
Drake

SOLUTION

MySQL / SQL Server

```sql
-- Check existence in both years efficiently
SELECT name FROM artists a
WHERE EXISTS (
    SELECT 1 FROM albums WHERE id=a.id AND yr=2024
)
AND EXISTS (
    SELECT 1 FROM albums WHERE id=a.id AND yr=2023
);
```

Next: Peak Revenue →

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SAVE REPOST

Problem 6/20 RANK()

UBER

Peak Revenue

Find the highest revenue trip per city using window functions.

SAMPLE INPUT

city	rev
NYC	500
NYC	300
LA	450

EXPECTED OUTPUT

city	rev	r
NYC	500	1
LA	450	1

SOLUTION

MySQL / SQL Server

```sql
-- Rank trips by revenue partitioned by city
SELECT city, rev, r FROM (
    SELECT *,
    RANK() OVER(PARTITION BY city ORDER BY rev DESC) as r
    FROM trips
) t
WHERE r = 1;
```

Next: Weekend Surge →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 7/20 WEEKDAY

DOORDASH

Weekend Surge

Count total orders placed on Saturday or Sunday.

SAMPLE INPUT

id	day_index
501	5 (Sat)
502	6 (Sun)
503	0 (Mon)

EXPECTED OUTPUT

weekend_count
2

SOLUTION

MySQL

```sql
-- Filter for Sat (5) or Sun (6)
SELECT COUNT(*) as weekend_count
FROM orders
WHERE WEEKDAY(order_date) IN (5, 6);
```

Next: Churn Logic →

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The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 8/20 NOT IN

NETFLIX

Churn Logic

Users active in Jan who did NOT return in Feb.

SAMPLE INPUT

u_id	month
10	Jan
10	Feb
11	Jan

EXPECTED OUTPUT

user_id
11

SOLUTION

MySQL / SQL Server

```sql
-- Exclude users present in month 2
SELECT user_id
FROM logs
WHERE month = 1
AND user_id NOT IN (
    SELECT user_id FROM logs WHERE month = 2
);
```

Next: Moving Average →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 9/20 | OVER(ROWS) | TIKTOK

Moving Average

Calculate a 3-day rolling average of views.

SAMPLE INPUT

date	views
01-01	10
01-02	20
01-03	30

EXPECTED OUTPUT

date	roll_avg
01-02	15
01-03	20

SOLUTION

MySQL / SQL Server

```sql
-- Avg of current row + 2 previous rows
SELECT
    date,
    AVG(views) OVER(
        ORDER BY date
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
    ) as roll_avg
FROM daily_stats;
```

Next: Revenue MoM →

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The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 10/20 | LAG()

STRIPE

Revenue MoM

Compare current month revenue with previous month.

SAMPLE INPUT

month	revenue
Jan	4000
Feb	5000

EXPECTED OUTPUT

month	revenue	prev_rev
Jan	4000	NULL
Feb	5000	4000

SOLUTION

MySQL / SQL Server

```sql
-- Access previous row's value
SELECT
    month, revenue,
    LAG(revenue) OVER(ORDER BY month) as prev_rev
FROM monthly_sales;
```

Next: User Cohorts →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 11/20

DATE_FORMAT

AIRBNB

User Cohorts

Group users by their signup month cohort.

SAMPLE INPUT

u_id	signup
1	2024-01-15
2	2024-01-20
3	2024-02-01

EXPECTED OUTPUT

cohort	user_count
2024-01-01	2
2024-02-01	1

SOLUTION

MySQL

```sql
-- Truncate date to first day of month
SELECT
    DATE_FORMAT(signup, '%Y-%m-01') as cohort,
    COUNT(*) as user_count
FROM users
GROUP BY 1;
```

Next: Stock Alerts →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE

REPOST

Problem 12/20 | CASE WHEN | WALMART

Stock Alerts

Flag products with stock < 10 as "Critical".

SAMPLE INPUT

item	qty
Apple	5
Banana	50

EXPECTED OUTPUT

item	status
Apple	Critical
Banana	Healthy

SOLUTION

MySQL / SQL Server

```sql
-- Conditional logic for status
SELECT
    item,
    CASE
        WHEN qty < 10 THEN 'Critical'
        ELSE 'Healthy'
    END as status
FROM inventory;
```

Next: Top 3 Per User →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 13/20 ROW_NUMBER APPLE

Top 3 Per User

Get the 3 most recent orders for every single user.

SAMPLE INPUT

u_id	order_dt
1	2024-05-01
1	2024-04-01
1	2024-01-01

EXPECTED OUTPUT

u_id	order_dt	rn
1	2024-05-01	1
1	2024-04-01	2

SOLUTION

MySQL / SQL Server

```sql
-- Assign unique rank per user order
SELECT u_id, order_dt, rn FROM (
    SELECT *,
    ROW_NUMBER() OVER(
        PARTITION BY u_id ORDER BY order_dt DESC
    ) as rn
    FROM orders
) t
WHERE rn ≤ 3;
```

Next: Org Hierarchy →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 14/20 RECURSIVE

MICROSOFT

Org Hierarchy

Find all employees reporting up to a specific manager.

SAMPLE INPUT

emp	mgr
Alice	NULL
Bob	Alice
Charlie	Bob

EXPECTED OUTPUT

id	name
2	Bob
3	Charlie

SOLUTION

MySQL / SQL Server

```sql
-- Recursive CTE for tree traversal
WITH RECURSIVE org AS (
    SELECT id, name FROM employees WHERE id=1
    UNION ALL
    SELECT e.id, e.name FROM employees e
    JOIN org o ON e.mgr_id = o.id
)
SELECT id, name FROM org;
```

Next: Login Streaks →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 15/20

LEAD()

TESLA

Login Streaks

Find consecutive days where a user logged in.

SAMPLE INPUT

u_id	date
1	Jan 1
1	Jan 2
1	Jan 5

EXPECTED OUTPUT

date	next
Jan 1	Jan 2

SOLUTION

SQL Server

```sql
-- Look ahead to next login date
SELECT date, next FROM (
    SELECT *,
    LEAD(date) OVER(PARTITION BY u_id ORDER BY date) as next
    FROM logs
) t
WHERE DATEDIFF(day, date, next) = 1;
```

Next: Null Cleanup →

Anandnarayanan S

The 20-Pattern SQL Cheat Sheet

SAVE

REPOST

Problem 16/20 COALESCE

IBM

Null Cleanup

Replace NULL phone numbers with "N/A".

SAMPLE INPUT

name	phone
Bob	555-0199
Alice	NULL

EXPECTED OUTPUT

name	contact
Bob	555-0199
Alice	N/A

SOLUTION

MySQL / SQL Server

```sql
-- Return first non-null value
SELECT
    name,
    COALESCE(phone, 'N/A') as contact
FROM customers;
```

Next: Power Users →

Anandnarayanan S

The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 17/20 HAVING

LINKEDIN

Power Users

Find users who have placed more than 10 orders.

SAMPLE INPUT

u_id	orders
55	12
56	4

EXPECTED OUTPUT

u_id

SOLUTION

MySQL / SQL Server

```sql
-- Filter aggregated groups
SELECT u_id
FROM orders
GROUP BY u_id
HAVING COUNT(*) > 10;
```

Next: Running Total →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 18/20

SUM() OVER

SALESFORCE

Running Total

Calculate cumulative revenue over time.

SAMPLE INPUT

date	amt
Jan	10
Feb	20
Mar	30

EXPECTED OUTPUT

date	amt	cumul
Jan	10	10
Feb	20	30

SOLUTION

MySQL / SQL Server

```sql
-- Rolling sum ordered by date
SELECT
    date, amt,
    SUM(amt) OVER(ORDER BY date) as cumul
FROM sales;
```

Next: Unshipped →

Anandnarayanan S

The 20-Pattern SQL Cheat Sheet

SAVE

REPOST

Problem 19/20 | IS NULL | PAYPAL

Unshipped

Find orders that have no delivery date set.

SAMPLE INPUT

order_id	del_date
101	2024-01-01
102	NULL

EXPECTED OUTPUT

order_id	status
102	Pending

SOLUTION

MySQL / SQL Server

```sql
-- Check for missing values
SELECT order_id, 'Pending' as status
FROM orders
WHERE del_date IS NULL;
```

Next: Find Duplicates →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

Problem 20/20 | GROUP BY | SNOWFLAKE

Find Duplicates

Find users with duplicate email registrations.

SAMPLE INPUT

email
a@co.com
a@co.com
b@co.com

EXPECTED OUTPUT

email	cnt
a@co.com	2

SOLUTION

MySQL / SQL Server

```sql
-- Group by email and count > 1
SELECT email, COUNT(*) as cnt
FROM users
GROUP BY email
HAVING COUNT(*) > 1;
```

Next: MISSION ACCOMPLISHED →

Anandnarayanan S
The 20-Pattern SQL Cheat Sheet

SAVE REPOST

MISSION ACCOMPLISHED

You're Interview Ready!

20 PATTERNS MASTERED

- ✓ DATE_FILTER
- ✓ DATEDIFF
- ✓ INNER JOIN
- ✓ DATE_FORMAT
- ✓ EXISTS
- ✓ RANK()
- ✓ WEEKDAY
- ✓ NOT IN
- ✓ OVER(ROWS)
- ✓ LAG()
- ✓ DATE_FORMAT
- ✓ CASE WHEN
- ✓ ROW_NUMBER
- ✓ RECURSIVE
- ✓ LEAD()
- ✓ COALESCE
- ✓ HAVING
- ✓ SUM() OVER
- ✓ IS NULL
- ✓ GROUP BY
- ★ FULL DECK

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